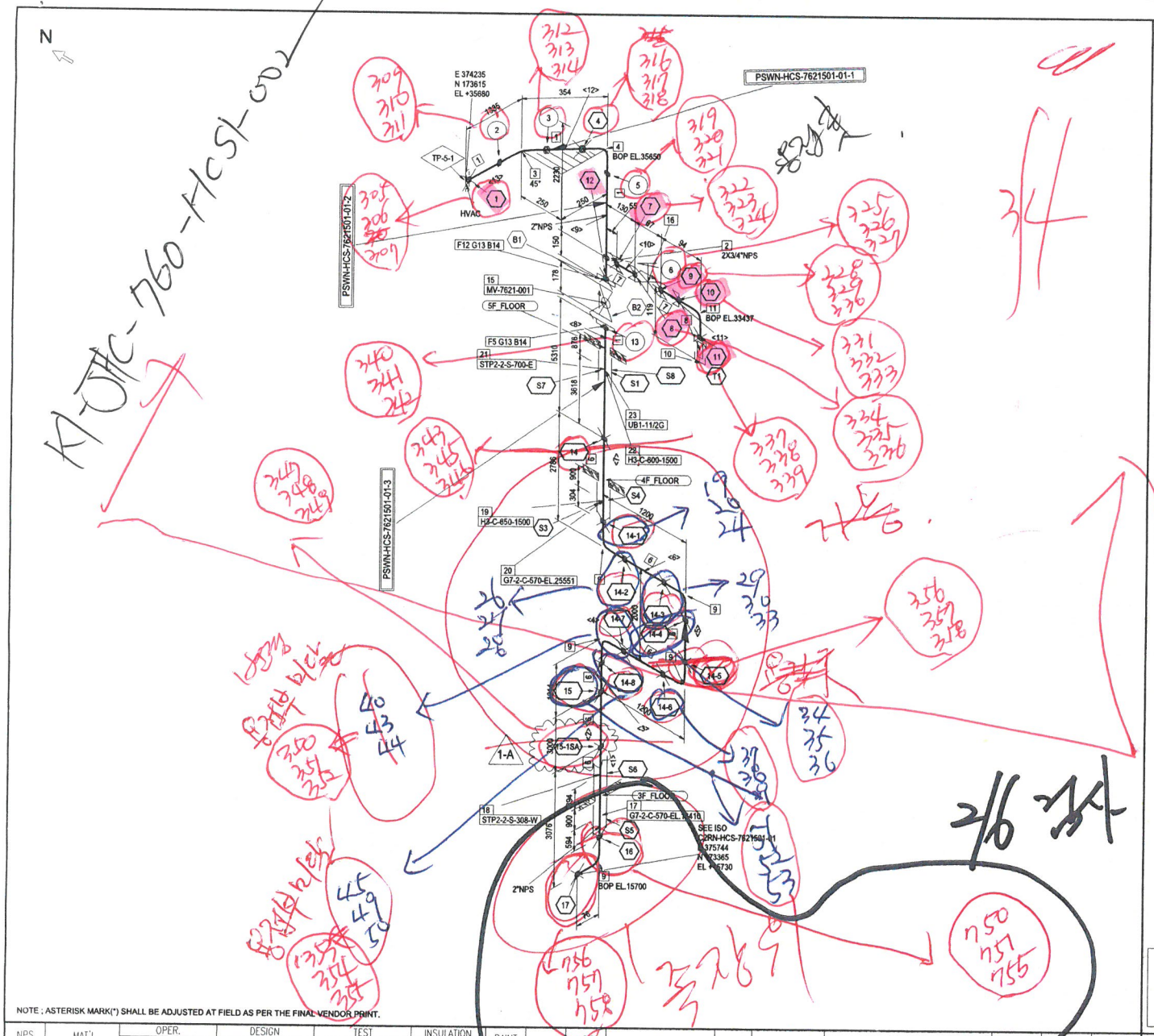

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°C)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°C)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
2	S98	3	144	4.0 / F.V	172	6	H	IHI	25	-	N	5	-
0.75	S98	3	144	4.0 / F.V	172	6	H	IHI	25		N	5	-

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
1-A	14-01-26		H.S.JUNG	M.H.SUNG	T.H.LIM
1	10.09.25	ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
0	24-06-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

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BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2"	9.2 M	S98	P	PA02-000001902
2	HALF COUPLING SW CL3000 ASTM A182-F316/316L	2X3/4"	1	S98	TH	PB99-000038951
3	45 DEG. ELBOW BW WELDED ASTM A403-WP316/316L SCH10S	2"	1	S98	EA	PB99-000038361
4	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	2"	1	S98	E	PB99-000038409
5	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	2"	1	S98	F	PB01-000023955

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
6	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2"	13.8 M	S98	P	PA02-000001902
7	PIPE SMLS ASTM A312-316/316L SCH40S PE	3/4"	0.2 M	S98	P	PA02-000000856
8	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
9	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	2"	5	S98	E	PB99-000038409
10	CAP SCRD (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
11	90 DEG. ELBOW (LR) SW CL3000 ASTM A182-F316/316L	3/4"	1	S98	E	PB99-000038441
12	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	2"	1	S98	F	PB01-000023955
13	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHITE 4.5MM(0.18") THK.	2"	2	S98	G	PF01-000008267
14	90 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNUNR x MM	5/8"	8	S98	B	PB99-000078892
15	BALL VALVE FLGD-RF CL150 ASTM A351-CF8M SS316 TRIM PTFE SEAT FLOATING BALL FB LV. OP. (VAB160F)	2"	1	S98	VL	PP04-000007540
16	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443
17	G7-2-C-570-EL.16410	2"	1			G7-C
18	STP2-2-S-308-W	2"	1			STP2
19	H3-C-650-1500	2"	1			H3-C
20	G7-2-C-570-EL.25551	2"	1			G7-C
21	STP2-2-S-700-E	2"	1			STP2
22	H3-C-600-1500	2"	1			H3-C
23	UB1-11/2G	1.1/2"	1			UB1-G

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	3000	2		<3>	3000	2	
<3>	1138	2		<4>	1048	2	
<5>	1848	2		<6>	1048	2	
<7>	2710	2		<8>	5300	2	
<9>	2294	2		<10>	75	3/4	
<11>	75	3/4		<12>	246	2	
<13>	1293	2					

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2026-01-14

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	PSWN
P&ID NO.	K1-DSL-UT-PID-7621-01			PROJ. NO.	16480	FLUID	HCS
GAD NO.	-			LINE NO.	2-HCS-7621501-S98-IHI		
DWG NO.	PSWN-HCS-7621501-01					SH'T NO. 1 OF 1	REV. 1-A

PIPE modified at 10:37:26 Aug 2025, Drawing Created at 11:09:15 28 August 2025
PIPE NAME : 2-HCS-7621501-S98-IHI_PSWN-01